

4.2 Double Integrals Worksheet

1. In the xy -plane, the region R is bounded by $y = \sqrt{x}$, $x = 4$ and the x -axis. Determine the volume of the solid bounded above by $f(x, y) = \frac{y}{1+x^2}$ and below by the region R .
2. In the xy -plane, the region R is bounded by the curves $y = x + 2$, $y = 2 - x$ and $x = -2$. Sketch the region and determine the volume of the solid bounded above by $f(x, y) = x^2$.

3. Use a double integral to compute the area of the region bounded by the graphs of $f(x) = \sin(x)$ and $g(x) = \cos(x)$ between $x = \frac{\pi}{4}$ and $x = \frac{5\pi}{4}$.

4. Use a double integral to determine the area of the region D bounded by $y = \ln(x)$, $y = e^x$ on the interval $1 \leq x \leq 2$.

5. Evaluate $\iint_D (2x - y) dA$ where D is the region bounded by circle centered at the origin with radius 2.

6. Evaluate $\iint_D y dA$ where D is the region bounded by $y^3 - 3y - x = 0$ and $y = x$.

7. Sketch the region of integration and change the order of integration.

a) $\int_0^2 \int_{x^2}^4 f(x, y) dy dx$

b) $\int_0^{\pi/2} \int_0^{\cos(x)} f(x, y) dy dx$

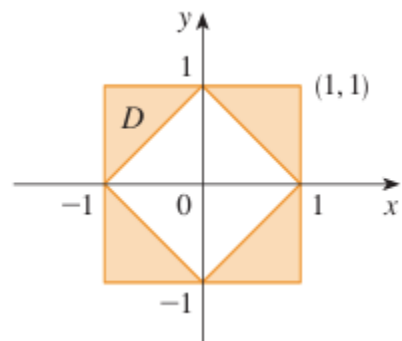
c) $\int_1^2 \int_0^{\ln(x)} f(x, y) dy dx$

8. Evaluate the double integral $\int_0^1 \int_x^1 e^{x/y} dy dx$.

9. Evaluate the double integral $\int_0^4 \int_{\sqrt{x}}^2 \frac{1}{y^3 + 1} dy dx$.

10. Express D as a union of regions and evaluate the integral.

a) $\iint_D x^2 dA$



b) $\iint_D y \, dA$

